FLAT, FINITE LOCALLY FREE AND FINITE FREE MODULES OVER A DEDEKIND DOMAIN

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Being torsion free is a local condition on the ring. Lemma \ref{lemma-check-torsion} and Algebra, Lemma \ref{algebra-lemma-flat-localization}, cited at proof lines 4966-4968. Reduce the equivalence of flatness and torsion freeness to the local rings.
- A localisation of a Dedekind domain at a maximal ideal is a discrete valuation ring, and over it flat is the same as torsion free. Algebra, Lemma \ref{algebra-lemma-characterize-Dedekind} and Lemma \ref{lemma-valuation-ring-torsion-free-flat}, cited at proof lines 4969-4971. Give the local statement that the reduction lands on.
- A module is finite projective if and only if it is finitely presented and flat. Algebra, Lemma \ref{algebra-lemma-finite-projective}, cited at proof lines 4974-4975. Turns flatness plus finiteness into finite local freeness.
- A finite torsion free module over a principal ideal domain embeds into a finite free module. Lemma \ref{lemma-finite-torsion-free-submodule-free}, cited at proof lines 4980-4981. Starts the induction for the third statement.

Source reference. Original lemma 0AUW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13664–13664. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0AUQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13658–13658. Navigation pointer to a statement the

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Source reference. Original lemma 0AUT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13661–13661. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 655–655. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 034X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4108–4108. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0539: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6597–6597. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 875–875. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0AUU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13662–13662. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be a Dedekind domain (for example a discrete valuation ring or more generally a PID).*

(1) *An A-module is flat if and only if it is torsion free.*
(2) *A finite torsion free A-module is finite locally free.*
(3) *A finite torsion free A-module is finite free if A is a PID.*

Proof. (For the parenthetical remark in the statement of the lemma, see Algebra, Lemma 0AUQ.) Proof of (1). By Lemma 0AUT and Algebra, Lemma 00HT it suffices to check the statement over $A_{\mathfrak{m}}$ for $\mathfrak{m} \subset A$ maximal. Since $A_{\mathfrak{m}}$ is a discrete valuation ring (Algebra, Lemma 034X) we win by Lemma 0539.

Proof of (2). Follows from Algebra, Lemma 00NX and (1).

Proof of (3). Let A be a PID and let M be a finite torsion free module. By Lemma 0AUU we see that $M \subset A^{\oplus n}$ for some n. We argue that M is free by induction on n. The case $n = 1$ expresses exactly the fact that A is a PID. If $n > 1$ let $M' \subset A^{\oplus n-1}$ be the image of the projection onto the last $n-1$ summands of $A^{\oplus n}$. Then we obtain a short exact sequence $0 \to I \to M \to M' \to 0$ where I is the intersection of M with the first summand A of $A^{\oplus n}$. By induction we see that M is an extension of finite free A-modules, whence finite free. □